Finding report: When graded PrudentBots cooperate with graded FairBots

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1 Introduction

Setting

Two programs play a one-shot prisoner's dilemma, and each can read the other's source code. A *modal agent* [1] decides by asking what Peano arithmetic (PA) can prove about the opponent, written with the provability box $\Box$ of the logic GL. Here p means "the opponent cooperates with me" and q means "the opponent cooperates with DefectBot", the agent that always defects. FairBot, $\Box p$, cooperates exactly when it can prove that the opponent cooperates with it. PrudentBot adds a safety test: it also requires a proof, in the stronger system PA+1, that the opponent would not cooperate with DefectBot, so it does not reward an opponent that cooperates with everyone.

We study two graded families. The graded FairBot $F_k = \Box^k p$ $(k \geq 1)$ asks for k nested layers of provability; F_1 is FairBot. The graded PrudentBot $P_j = \Box p \wedge \Box(\neg\Box^j \bot \to \neg q)$ $(j \geq 0)$ runs its safety test in PA+j, since $\neg\Box^j \bot$ expresses the consistency of PA+$(j-1)$; P_1 is PrudentBot.

Result

P_j and F_k cooperate with each other exactly when $k \leq j$; when $k > j$, both defect. In words: a PrudentBot that runs its safety test in a stronger system can afford to cooperate with FairBots that demand more nested proof, and the exchange rate is one for one. Every P_j still defects against CooperateBot, so the extra cooperation costs nothing in safety.

Why it matters

Barasz et al. remark that PrudentBot's safety test may use a stronger system than PA+1 and that this widens the set of agents it cooperates with, and online discussion of the paper looked at specific pairs of such variants. Here the relation is pinned down exactly for the two graded families, for all indices: the boundary of cooperation is the single inequality $k \leq j$.

Proved vs checked

Proved for all $j \geq 0$ and $k \geq 1$ in Lean 4, in the height semantics of closed GL fixed points (on the chain of heights, both agents cooperate at every height when $k \leq j$; when $k > j$ both defect above height $j + k$). The link to provability in PA is Solovay's theorem [3] (see also [2]), cited, not formalised, and the reading in the standard model assumes PA is sound.

Novelty statement

This result extends Barasz et al. (arXiv 1401.5577), remark after Theorem 3.2: PrudentBot's DefectBot check may use PA+n for larger n; LessWrong comments on "Robust Cooperation in the Prisoner's Dilemma" (orthonormal, cousin_it): indexed Prudent(m,n)Bot variants with specific pairs. Our prior-art search (2026-10-21) covered:

- Barasz et al. 2014, Robust Cooperation in the Prisoner's Dilemma (arXiv 1401.5577), full text: agents CB, DB, CliqueBot, FairBot, PrudentBot, TrollBot, JustBot, WaitFairBot_K; Theorems 3.2, 4.8, 4.10 and the obstacles section.
- LaVictoire et al. 2014, Program Equilibrium in the Prisoner's Dilemma via Löb's Theorem (intelligence.org/files/ProgramEquilibrium.pdf).
- LessWrong post "Robust Cooperation in the Prisoner's Dilemma" and its comments (orthonormal, cousin_it, Will_Sawin, Karl, dankane).
- Arbital / LessWrong wiki "Modal combat".
- Garrabrant, "Prisoners' Dilemma with Costs to Modeling" (Alignment Forum).
- Mennen, "Modal Combat for games other than the prisoner's dilemma" (agentfoundations.org item 1304).
- github.com/machine-intelligence/provability (Haskell solver and combat code; the agents folder contents were not visible).

If you know of related work we missed, please open an issue.

How it was produced

This result was found by AI research agents (the autoresearch harness). It was checked by scripts that run Lean proofs; the proof and check files are included so anyone can re-run them.

2 Claims

Claim K8

`For all j>=0, k>=1 (height-chain semantics of closed GL fixed points): P_j=[]p&[](~[]^jF->~q) and F_k=[]^k p cooperate at every height iff k<=j; if j<k both defect above height j+k. Lean-proved; P_j defects vs CB.`

Status: derived ů author check only.

Verified.

- `checks/K8-c046_pjfk_lean.py`: embedded Lean 4 proof of pj_fk and pj_cb compiles with no sorryAx; flipped-conclusion variant is rejected by Lean; explicit height table for j<=8,k<=8 matches (coop everywhere iff k<=j, else both defect above j+k with exact threshold) — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: Closed (letter-free) GL fixed points are decided by height on finite linear frames (Boolos, Logic of Provability, ch. 7), so box phi at height h means phi at all g\<h; the Lean theorem is stated in exactly this semantics for arbitrary solutions.

Prior art. Partial prior art.

Provenance. Run `20261017-1126`, claims K8, run-folder commit `662166ecff0651902cc5fcb04bf747c62d79` published 2026-10-18.

3 How to verify

Each finding has its check scripts in `claims/graded-prudent-fair/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

4 Revisions

- 2026-10-21 — Novelty statement: render the Related work list instead of the whole prior-art answer

References

[1] Mihaly Barasz, Paul Christiano, Benja Fallenstein, Marcello Herreshoff, Patrick LaVictoire, and Eliezer Yudkowsky. Robust cooperation in the prisoner's dilemma: Program equilibrium via provability logic, 2014.

[2] George Boolos. *The Logic of Provability*. Cambridge University Press, 1993.

[3] Robert M. Solovay. Provability interpretations of modal logic. *Israel Journal of Mathematics*, 25:287–304, 1976.